Pharmacologic Actions, Therapeutic Use, Side Effects, Indications, Contraindications, and Nursing Responsibilities (i.e., Analgesics, Narcotics, Corticosteroids, Antihypertensives, Vasopressors, Antibiotics, Parenteral Fluids)

AKO SI CY

Allina Health Pain Assessment Scale

10	**Worst Pain You Can Imagine**
7-9	**Severe Pain** **Pain keeps you from doing your regular activities.** ⑨ Pain is so bad that you can't do any of your regular activities, including talking or sleeping. ⑧ Pain is so intense that you have trouble talking. ⑦ Pain distracts you and limits your ability to sleep.
4-6	**Moderate Pain** **Pain may interfere with your regular activities.** ⑥ Pain makes it hard to concentrate. ⑤ You can't ignore the pain but you can still work through some activities. ④ You can ignore the pain at times.
1-3	**Mild Pain** **Pain doesn't interfere with your regular activities.** ③ You may notice the pain but you can tolerate it. ② You may feel some twinges of pain. ① You may barely notice the pain.
0	**No Pain**

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1. Analgesics (Non-Narcotic)

Examples:

Acetaminophen, NSAIDs (ibuprofen, aspirin)

- **Therapeutic Use**
  - Mild to moderate pain
  - Fever
  - Inflammation (NSAIDs)

- **Side Effects:**
  - GI irritation, ulcers (NSAIDs)
  - Bleeding risk (aspirin)
  - Hepatotoxicity (acetaminophen)

- **Contraindications:**
  - Liver disease (acetaminophen)
  - GI bleeding, kidney disease (NSAIDs)
  - Aspirin allergy

- **Nursing Responsibilities**
  - Assess pain level before and after administration
  - Monitor liver enzymes with acetaminophen
  - Give NSAIDs with food
  - Teach max daily dose (acetaminophen ≤ 4 g/day)

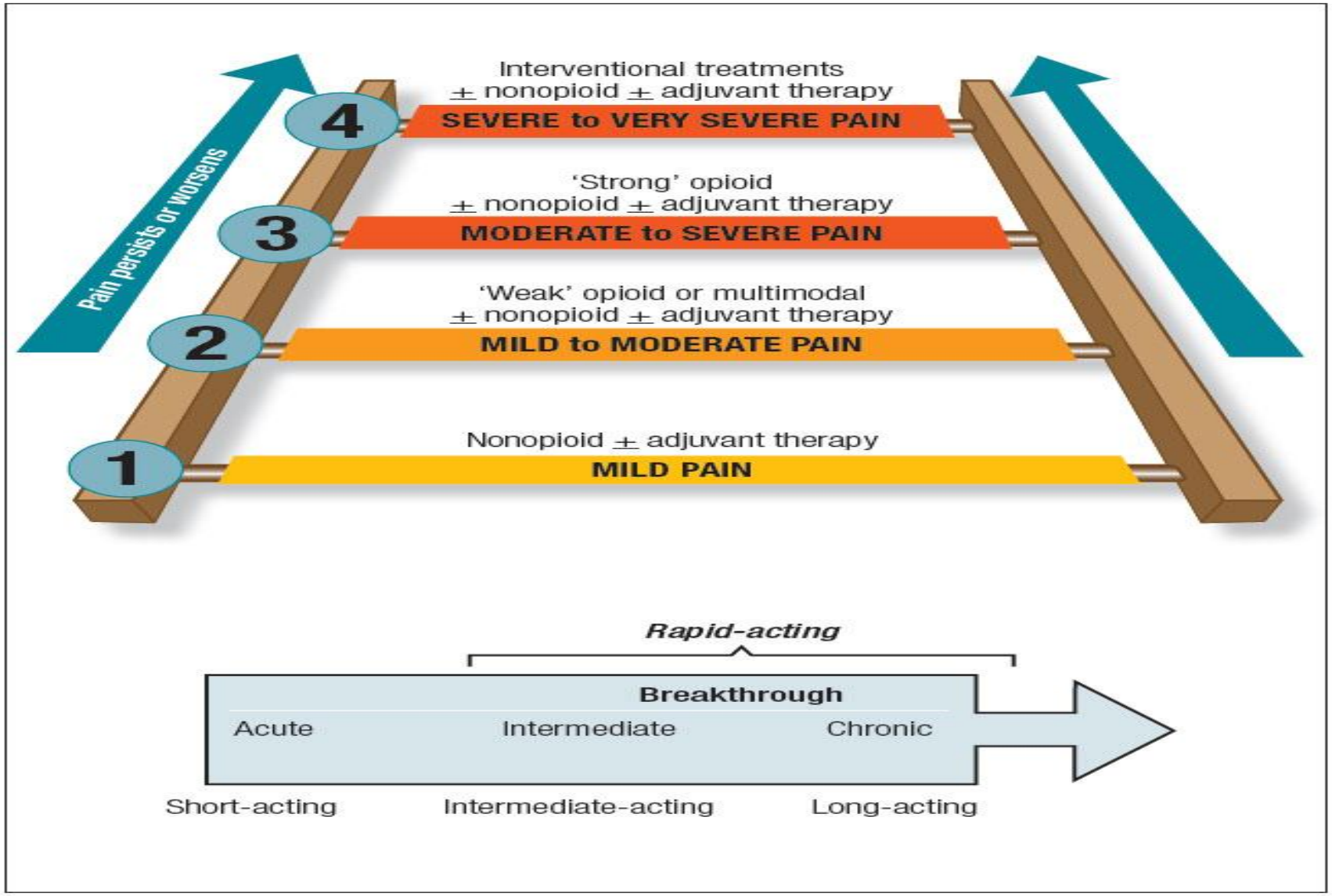


Figure 3. An updated version of the 1986 WHO pain ladder. Persistent and chronic pain syndromes should be treated with long-acting opioids; rapid-onset opioids are appropriate for breakthrough pain. A fourth step has been added for "very severe" pain that can be treated with peripheral nerve blockade.

- **2. Narcotics (Opioid Analgesics)**
  - **Examples:** Morphine, Fentanyl, Oxycodone

- **Therapeutic Use:**
  - Moderate to severe pain
  - Sedation

- **Side Effects:**
  - Respiratory depression
  - Sedation
  - Constipation
  - Nausea
  - Urinary retention

- **Indications:**
  - Post-op pain
  - Cancer pain
  - Severe trauma

- **Contraindications:**
  - Respiratory depression
  - Head injury (↑ ICP)
  - Opioid dependency (use caution)

- **Nursing Responsibilities:**
  - Assess **respiratory rate** (hold if <12/min)
  - Monitor LOC and pain scale
  - Have **naloxone** available
  - Implement fall precautions
  - Monitor bowel function

- **3. Corticosteroids:**
  - **Examples:** Prednisone, Dexamethasone, Hydrocortisone

- **Pharmacologic Action:**
  - Suppress inflammation and immune response
  - Mimic adrenal hormones

- **Therapeutic Use:**
  - Autoimmune disorders
  - Asthma/COPD
  - Allergic reactions
  - Shock

- **Side Effects:**
  - Hyperglycemia
  - Immunosuppression
  - Fluid retention
  - Osteoporosis
  - Weight gain

- **Indications:**
  - Inflammatory conditions
  - Adrenal insufficiency

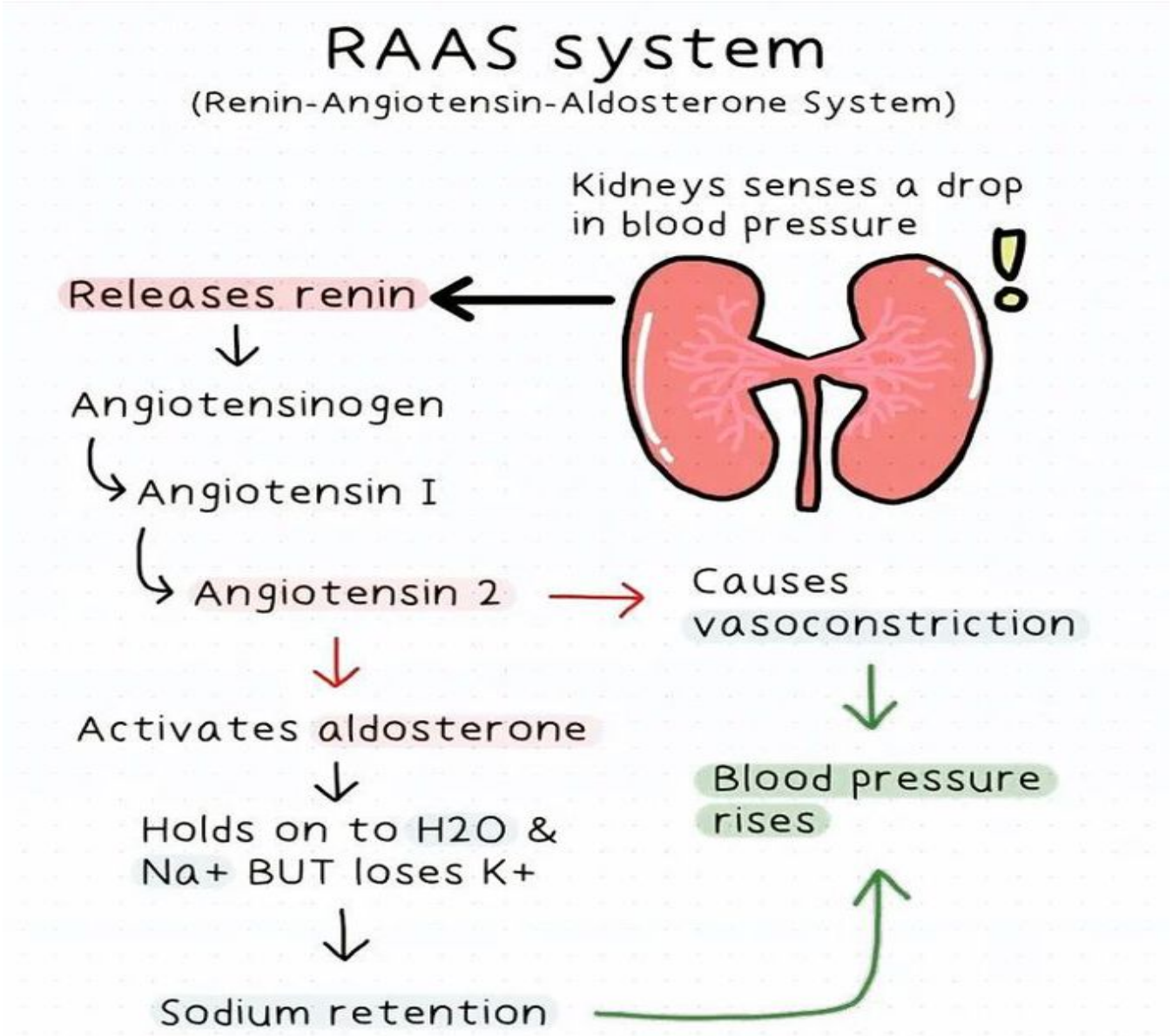
- **Contraindications:**
  - Systemic infections
  - Uncontrolled diabetes (caution)

- **Nursing Responsibilities:**
  - Monitor blood glucose
  - Watch for infection (masked symptoms)
  - Administer with food
  - **Never stop abruptly** (taper!)
  - Monitor electrolytes

- **4. Antihypertensives:**
  - **Examples:** Beta-blockers, ACE inhibitors, Calcium channel blockers

RAAS system

(Renin-Angiotensin-Aldosterone System)



- **Pharmacologic Action:**
  - Lower blood pressure by ↓ cardiac output or vasodilation

- **Therapeutic Use:**
  - Hypertension
  - Heart failure
  - Post-MI

- **Side Effects:**
  - Hypotension
  - Bradycardia
  - Dizziness
  - Cough (ACE inhibitors)

- **Indications:**
  - Elevated blood pressure

- **Contraindications:**
  - Hypotension
  - Bradycardia
  - Pregnancy (ACE inhibitors)

- **Nursing Responsibilities:**
  - Check BP & HR before administration
  - Monitor orthostatic hypotension
  - Educate the patient not to stop abruptly
  - Monitor potassium with ACE inhibitors

- **5. Vasopressors**
  - **Examples:** Norepinephrine, Dopamine, Epinephrine

- **Pharmacologic Action**
  - Vasoconstriction → ↑ blood pressure

- **Therapeutic Use:**
  - Shock
  - Severe hypotension

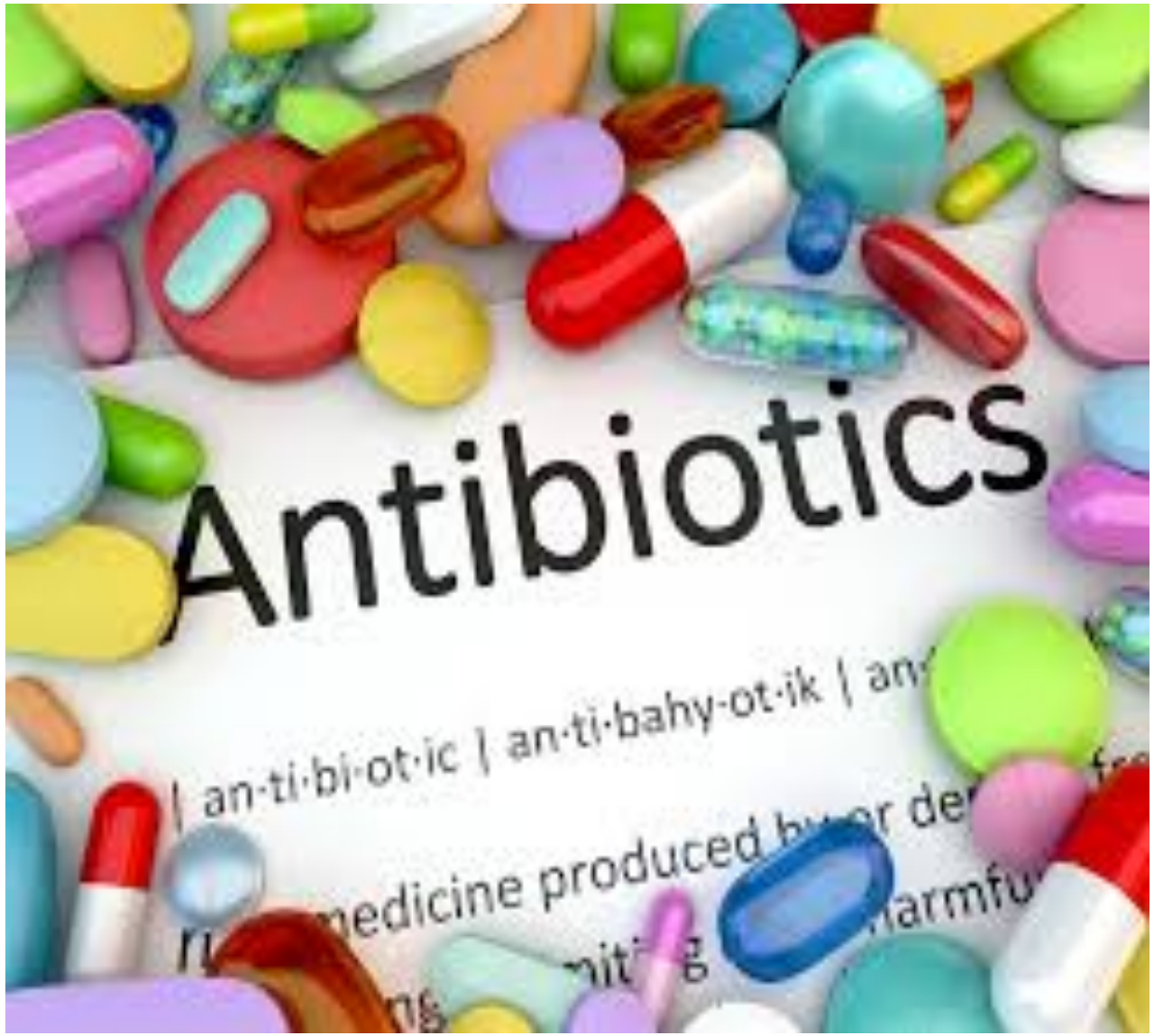
- **Side Effects:**
  - Hypertension
  - Arrhythmias
  - Tissue necrosis (extravasation)

- **Indications:**
  - Septic shock
  - Cardiogenic shock

- **Contraindications:**
  - Hypovolemia (correct fluids first)

- **Nursing Responsibilities:**
  - Continuous BP monitoring (arterial line preferred)
  - Administer via **central line**
  - Monitor IV site closely
  - Titrate per protocol
  - Monitor urine output

- **6. Antibiotics**
  - **Examples:** Penicillins, Cephalosporins, Vancomycin


Antibiotics
an·ti·bi·ot·ic | an·ti·bahy·ot·ik

- **Pharmacologic Action:**
  - Kill or inhibit bacterial growth

- **Therapeutic Use:**
  - Bacterial infections

- **Side Effects:**
  - Allergic reactions
  - GI upset
  - Nephrotoxicity (some classes)

- **Indications:**
  - Confirmed or suspected bacterial infection

- **Contraindications:**
  - Drug allergies
  - Severe renal impairment (dose adjusted)

- **Nursing Responsibilities:**
  - Check for allergies **before first dose**
  - Obtain cultures before starting
  - Monitor labs (BUN/Cr, trough levels)
  - Educate to complete the full course
  - Monitor for signs of superinfection

7. Parenteral Fluids

- **Examples:** NS, LR, D5W

IV FLUID TYPES



HOW IT WORKS:

- **Lower osmolarity** than the blood
- Causes **water to move from the extracellular space to the intracellular space**

TYPES:

- 0.45% Saline
- 0.225% Saline
- 0.33% Saline
- 5% Dextrose in Water*

*starts out as isotonic but turns hypotonic

RISKS:

CELL SWELLING & RUPTURE





HOW IT WORKS:

- **Equal osmolarity** as the blood
- Water **equally transfers in & out of the cell**
- Helps expand intravascular space

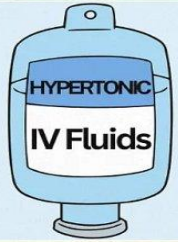
TYPES:

- Lactated Ringer's solution (LR)
- 5% Dextrose in water*
- 0.9% Normal Saline

ONLY solution used with **blood transfusions**

RISKS:

- Fluid **overload**
- **Increase** glucose (**D5W**)
- **LR**: not for pts with liver disease
  - can't convert lactate to bicarb



HOW IT WORKS:

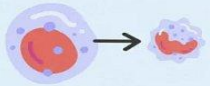
- **Higher osmolarity** than the blood
- Causes **water to move from the intracellular space to the extracellular space**

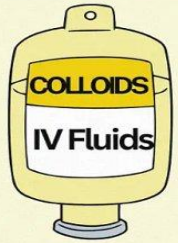
TYPES:

- 3% Saline
- Dextrose 10% in water
- Dextrose 5% in 0.9% Normal Saline
- Dextrose 5% in 0.45% Normal Saline

RISKS:

CELL DEHYDRATION & SHRINKAGE





HOW IT WORKS:

- **Large molecules** that **don't** cross capillary wall but **stay within intravascular space**
- **"Pulls"** water through **oncotic pressure from interstitial space to intravascular space**

TYPES:

- Albumin
- Dextran
- Hydroxyethyl starches (HES)
- Blood products
  - plasma, platelets, whole blood

RISKS:

- **Allergic** reactions
- Fluid volume **overload**
- **Bleeding** problems



©Nurse Sarah

- **Pharmacologic Action:**
  - Restore fluid and electrolyte balance
  - Expand intravascular volume

- **Therapeutic Use:**
  - Dehydration
  - Shock
  - Electrolyte imbalance

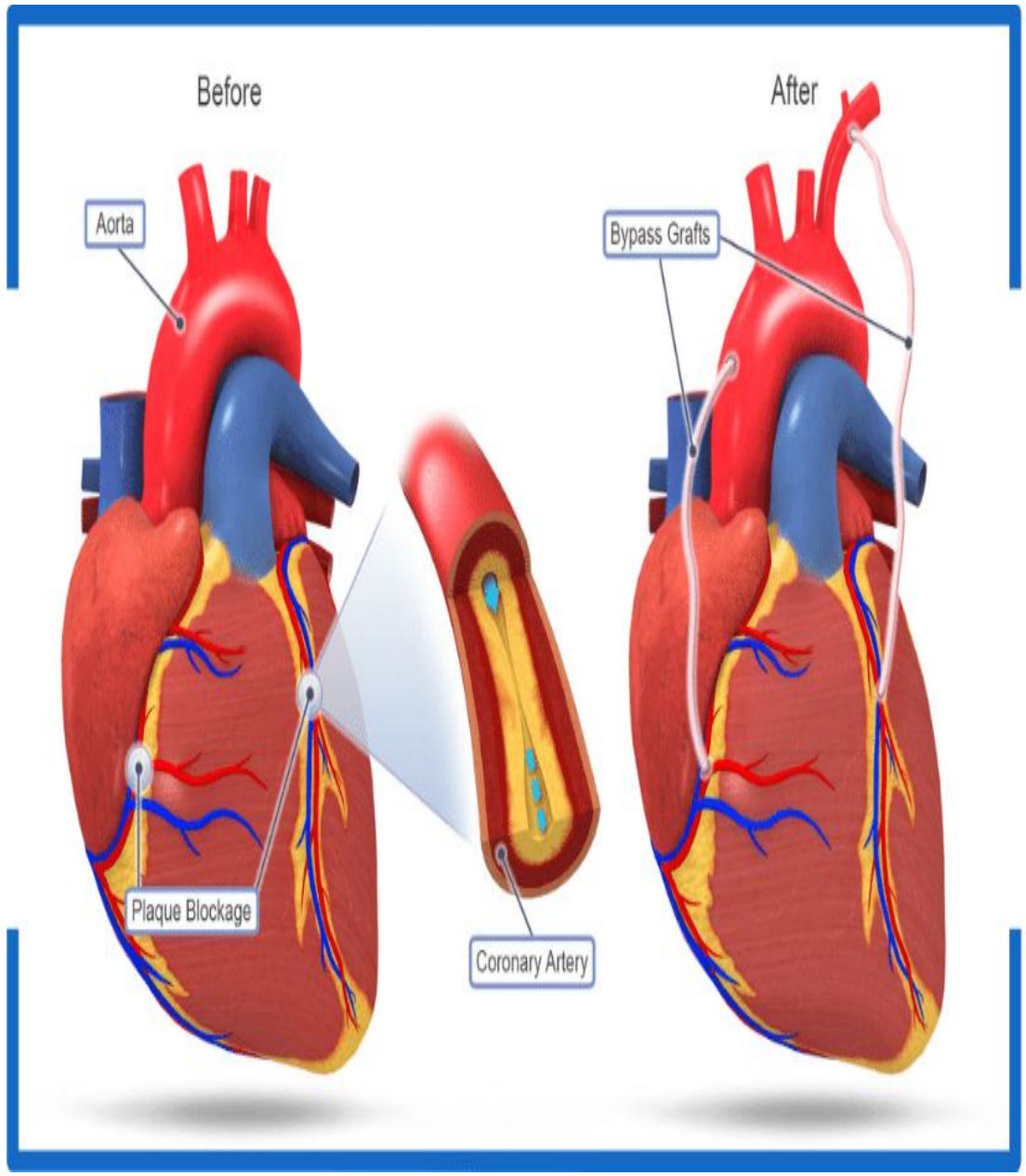
- **Side Effects:**
  - Fluid overload
  - Electrolyte imbalance
  - Pulmonary edema

- **Indications:**
  - Hypovolemia
  - NPO status

- **Contraindications:**
  - Heart failure
  - Renal failure (use cautiously)

- **Nursing Responsibilities:**
  - Monitor intake & output
  - Assess lung sounds and edema
  - Monitor electrolytes
  - Check IV site for infiltration
  - Regulate infusion rate carefully

- **A. Coronary Artery Bypass Graft (CABG)**
  - **Purpose:**
    - Restore blood flow to the myocardium by bypassing blocked coronary arteries



CABG

Coronary artery bypass surgery

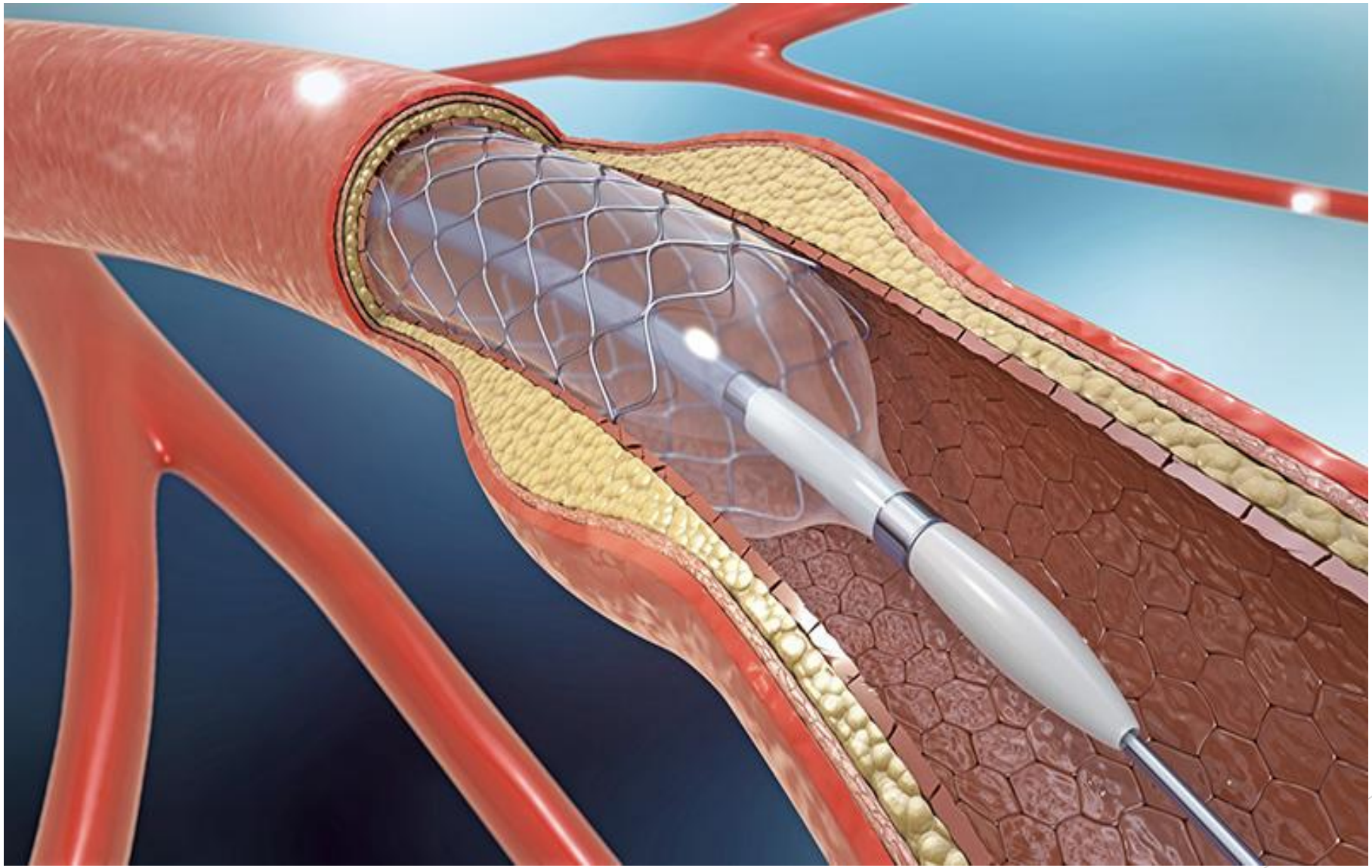
CABG is a heart bypass surgery, a medical procedure to improve blood flow to the heart.

- **Indications:**
  - Severe coronary artery disease
  - Multiple vessel blockage
  - Failed angioplasty or stent
  - Angina unresponsive to medical therapy

- **Nursing Responsibilities:**
  - **Pre-Op:**
    - Obtain informed consent
    - Assess cardiac status, labs, ECG
    - Teach deep breathing, coughing, and incentive spirometry
    - NPO status, pre-op skin prep

- **Post-Op:**
  - Monitor vital signs and ECG continuously
  - Assess chest tube drainage
  - Monitor for dysrhythmias
  - Assess incision sites
  - Manage pain and promote early ambulation
  - Monitor intake/output and electrolytes

- **B. Angioplasty (Percutaneous Coronary Intervention - PCI)**
  - **Purpose:**
    - Open narrowed coronary arteries using a balloon (often with stent placement)



- **Indications:**
  - Coronary artery stenosis
  - Acute myocardial infarction
  - Angina

- **Nursing Responsibilities:**
  - **Pre-Procedure:**
    - Check allergy to contrast dye
    - Assess renal function
    - Ensure informed consent
    - NPO as ordered

- **Post-Procedure:**
  - Monitor vital signs and ECG
  - Assess the puncture site for bleeding or hematoma
  - Keep the affected extremity straight
  - Monitor peripheral pulses
  - Encourage fluids to flush contrast
  - Observe for chest pain or dysrhythmias

- **C. Skin Grafting**
  - **Purpose**
    - Cover wounds and promote healing when skin loss is extensive



- **Autograft:**
  - Skin taken from the **same person**
  - **Best option** – no rejection

- **Allograft (Homograft):**
  - Skin from **another human**
  - Temporary cover (e.g., burn patients)

- **Xenograft (Heterograft):**
  - Skin from **another species** (usually pig)
  - Temporary biological dressing

- **Isograft:**
  - From a **genetically identical twin**

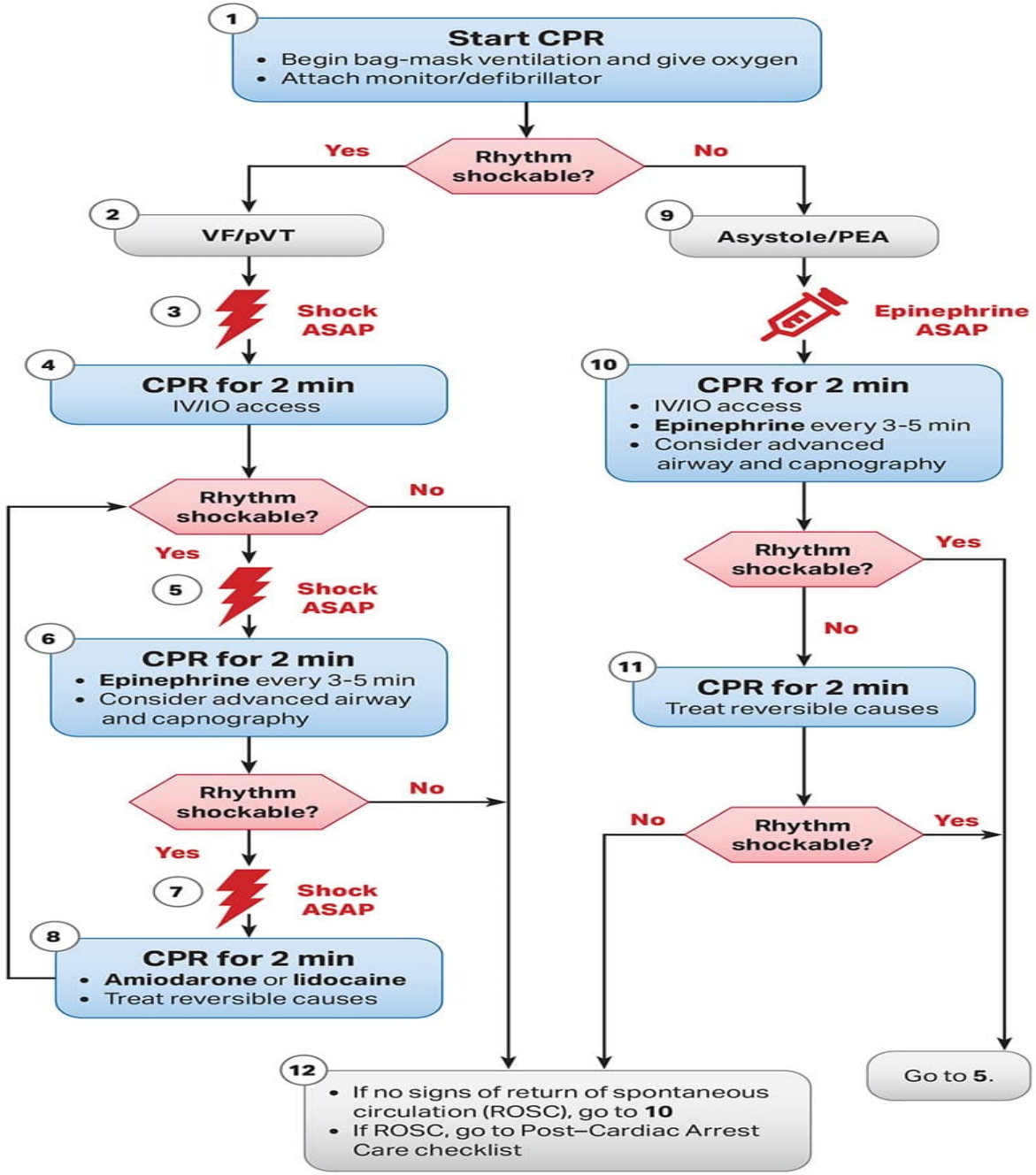
- **Indications**
  - Burns
  - Chronic ulcers
  - Traumatic skin loss

- **Nursing Responsibilities:**
  - **Pre-Op:**
    - Assess wound size and condition
    - Prepare donor and recipient sites
    - Manage pain

- **Post-Op:**
  - Immobilize the graft site
  - Monitor for infection
  - Assess graft viability (color, temperature)
  - Maintain moist dressings
  - Monitor donor site healing

- **A. Advanced Life Support (ALS)**
  - **Purpose:**
    - Sustain life during cardiac or respiratory arrest

Pediatric Cardiac Arrest Algorithm



High-Quality CPR

- Push hard (≥⅓ AP diameter of the chest)
- Push fast: 100-120/min
- Allow complete chest recoil
- Minimize interruptions in compressions
- Change compressor every 2 min, sooner if fatigued
- If no advanced airway, use compression-ventilation ratio of
  - 15:2 if 2 rescuers (prepuberty)
  - 30:2 if 2 rescuers (postpuberty onset)
  - 30:2 if 1 rescuer (any age)
- If advanced airway, provide continuous compressions and give a breath every 2-3 seconds
- Monitor $ETCO_2$ and, when available, invasive diastolic BP

Shock Energy for Defibrillation

- First shock 2 J/kg
- Second shock 4 J/kg
- Subsequent shocks ≥4 J/kg, maximum 10 J/kg or adult dose

Drug Therapy

- **Epinephrine IV/IO dose:** 0.01 mg/kg (0.1 mg/mL concentration). Max dose 1 mg.
- **Amiodarone IV/IO dose:** 5 mg/kg bolus (max 300 mg). May repeat up to 3 doses (max 150 mg subsequent doses).
  or
  Lidocaine IV/IO dose: 1 mg/kg

Advanced Airway

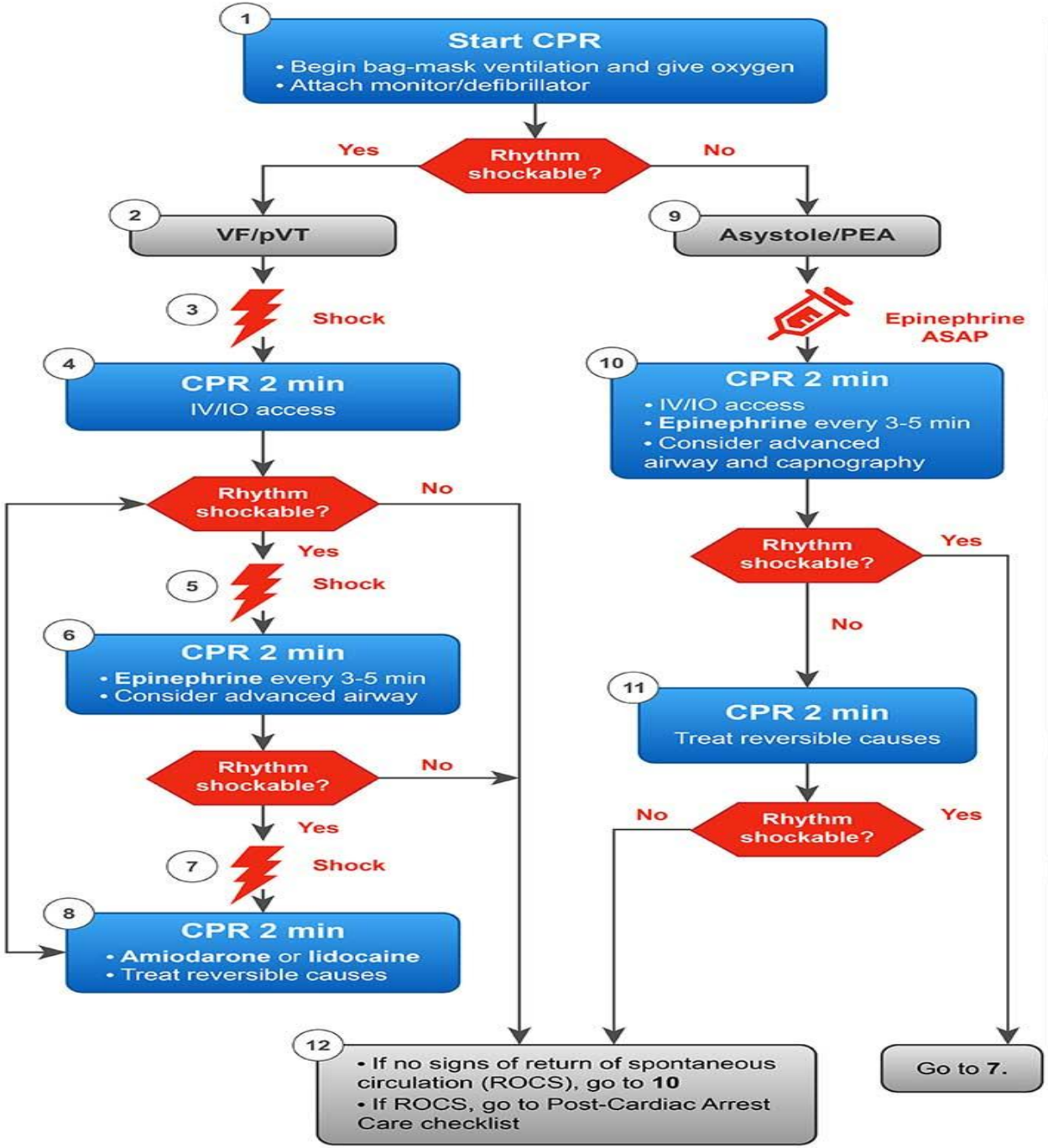
- ET intubation or supraglottic airway
- $ETCO_2$ to confirm and monitor ET tube placement

Reversible Causes

- **H**ypovolemia
- **H**ypoxia
- **H**ydrogen ion (acidosis)
- **H**ypoglycemia
- **H**ypo-/hyperkalemia
- **H**ypothermia
- **T**ension pneumothorax
- **T**amponade, cardiac
- **T**oxins
- **T**hrombosis, pulmonary
- **T**hrombosis, coronary

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Pediatric Cardiac Arrest Algorithm



CPR Quality

- Push hard (≥1/3 of anteroposterior diameter of chest) and fast (100-120/min) and allow complete chest recoil
- Minimize interruptions in compressions
- Change compressor every 2 minutes, or sooner if fatigued
- If no advanced airway, 15:2 compression-ventilation ratio
- If advanced airway, provide continuous compressions and give a breath every 2-3 seconds

Shock Energy for Defibrillation

- First shock 2 J/kg
- Second shock 4 J/kg
- Subsequent shocks ≥4 J/kg, maximum 10 J/kg or adult dose

Drug Therapy

- **Epinephrine IV/IO dose:** 0.01 mg/kg (0.1 mL/kg of the 0.1 mg/mL concentration). Max dose 1mg. Repeat every 3-5 minutes. If no IV/IO access, may give endotracheal dose: 0.1 mg/kg (0.1 mL/kg of the 1 mg/mL concentration).
- **Amiodarone IV/IO dose:** 5 mg/kg bolus during cardiac arrest. May repeat up to 3 total doses for refractory VF/pulseless VT

or

Lidocaine IV/IO dose: Initial: 1 mg/kg loading dose

Advanced Airway

- Endotracheal intubation or supraglottic advanced airway
- Waveform capnography or capnometry to confirm and monitor ET tube placement

Reversible Causes

- Hypovolemia
- Hypoxia
- Hydrogen ion (acidosis)
- Hypoglycemia
- Hypo-/hyperkalemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary

- **Indications:**
  - Cardiac arrest
  - Respiratory failure
  - Life-threatening arrhythmias

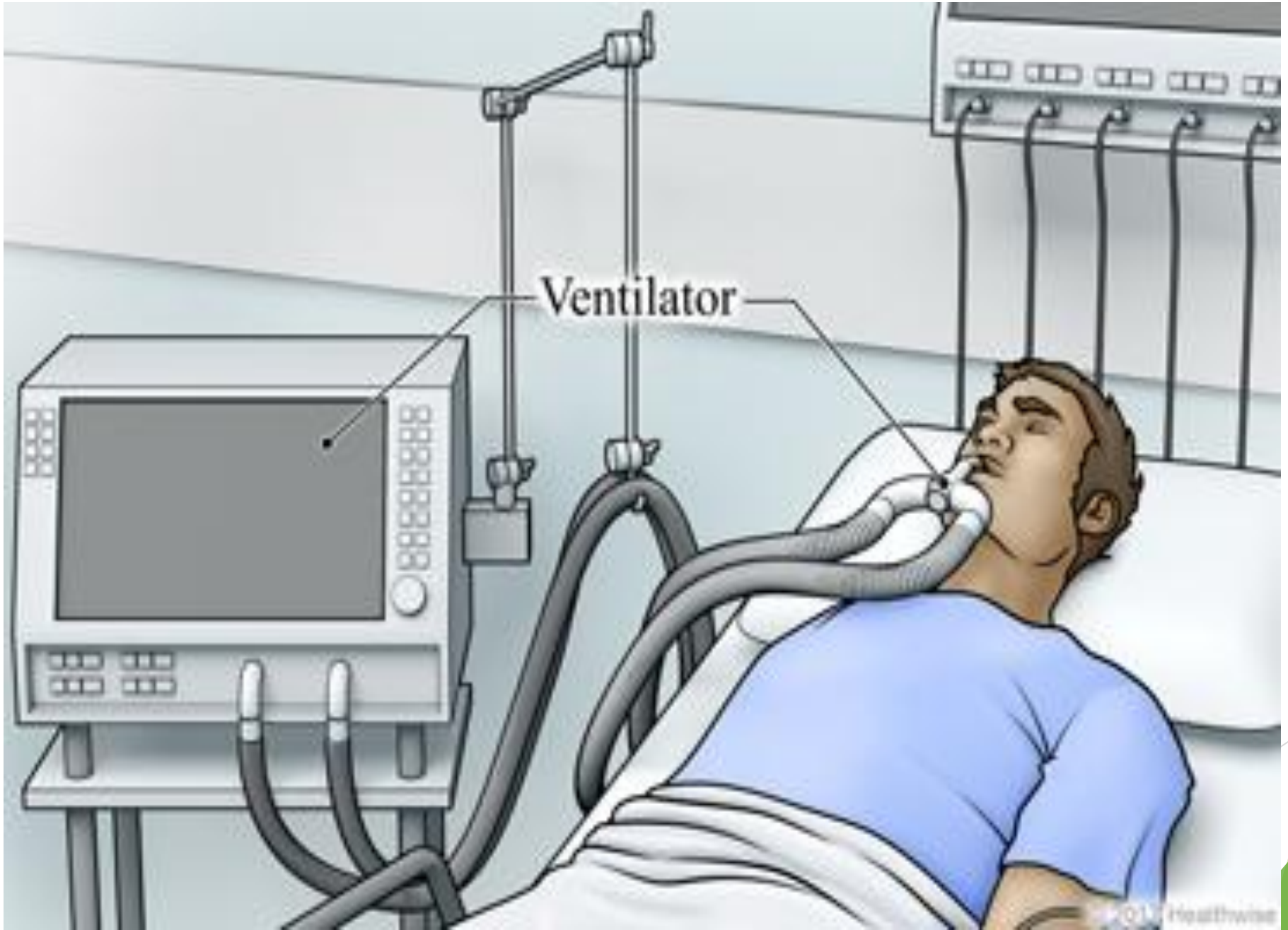
- **Nursing Responsibilities:**
  - Initiate CPR
  - Maintain airway and oxygenation
  - Assist with defibrillation
  - Administer emergency medications
  - Monitor cardiac rhythm
  - Document events accurately

- **B. Parenteral Hyperalimentation (Total Parenteral Nutrition - TPN)**
  - **Purpose:**
    - Provide complete nutrition intravenously

- **Indications:**
  - GI dysfunction
  - Severe malnutrition
  - NPO for extended periods

- **Nursing Responsibilities:**
  - Administer via **central line**
  - Use an infusion pump
  - Monitor blood glucose frequently
  - Change tubing and dressing per protocol
  - Monitor for infection and electrolyte imbalance
  - Never stop abruptly (risk of hypoglycemia)

- **C. Ventilatory Support**
  - **Purpose:**
    - Maintain adequate oxygenation and ventilation


Ventilator

- **Indications:**
  - Respiratory failure
  - Severe hypoxia
  - Decreased LOC

- **Nursing Responsibilities:**
  - Monitor ABGs and SpO_2
  - Ensure airway patency
  - Perform suctioning as needed
  - Prevent ventilator-associated pneumonia (VAP bundle)
  - Provide oral care
  - Assess readiness for weaning

- **D. Fluid Resuscitation:**
  - **Purpose:**
    - Restore circulating blood volume

- **Indications:**
  - Shock
  - Trauma
  - Severe dehydration

- **Nursing Responsibilities:**
  - Rapid IV access (large bore)
  - Monitor vital signs and urine output
  - Assess for fluid overload
  - Administer isotonic fluids as ordered
  - Monitor labs (electrolytes, lactate)

- **E. First Aid Measures (Emergency Wound & Trauma Management)**
  - **Purpose:**
    - Prevent further injury and stabilize patient

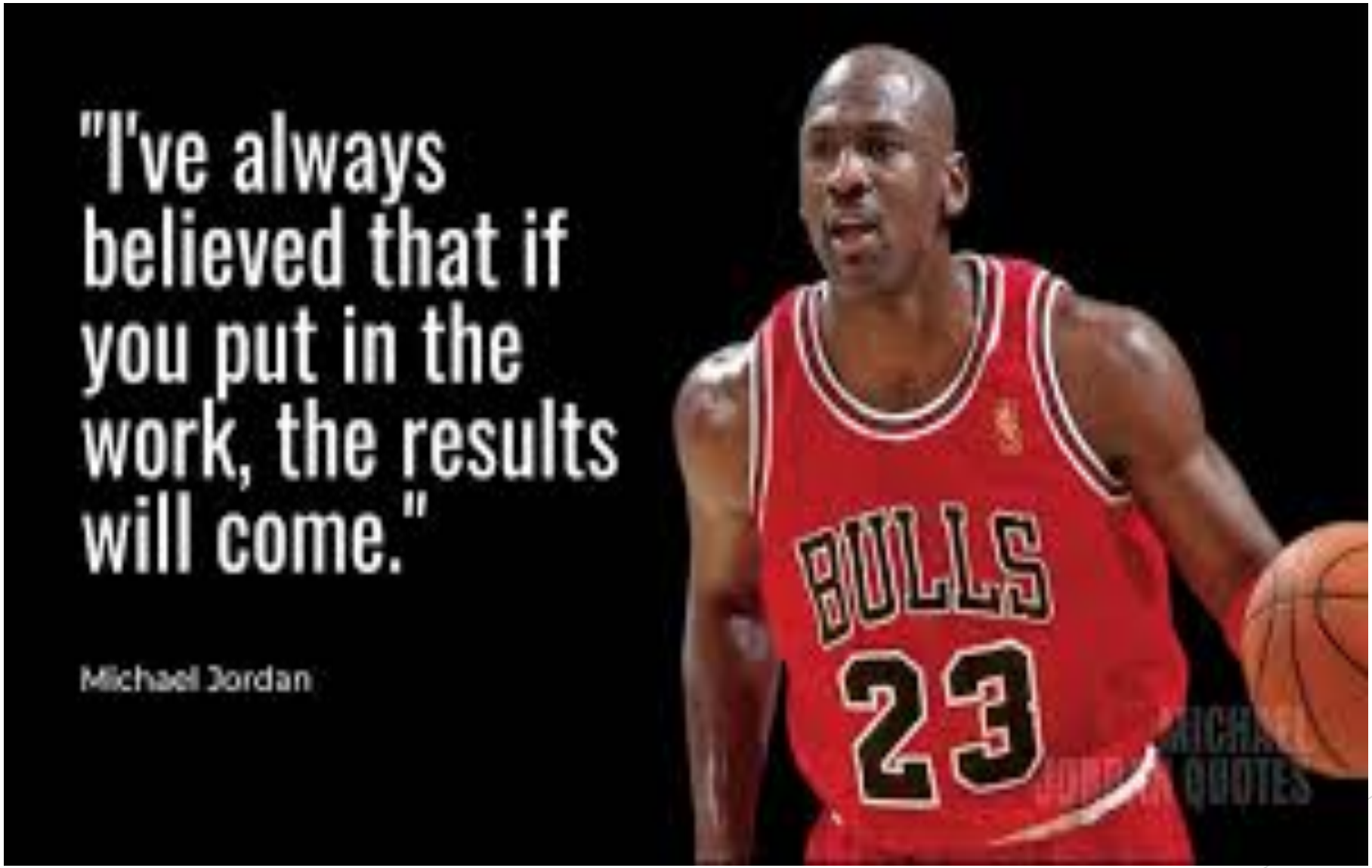
- **Indications:**
  - Trauma
  - Bleeding
  - Burns
  - Fractures

- **Nursing Responsibilities:**
  - Ensure scene safety
  - Control bleeding
  - Immobilize injuries
  - Clean and dress wounds
  - Monitor for shock
  - Provide tetanus prophylaxis as needed

- **F. IV Therapy**
  - **Purpose:**
    - Deliver fluids, medications, and blood products

- **Indications:**
  - Dehydration
  - Medication administration
  - Blood transfusion

- **Nursing Responsibilities:**
  - Select appropriate IV site and gauge
  - Maintain aseptic technique
  - Monitor IV site for infiltration or phlebitis
  - Regulate flow rate
  - Monitor intake/output
  - Educate patient on IV care


"I've always believed that if you put in the work, the results will come."
Michael Jordan
BULLS
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